

Two limit rules

Key Points

Two important limits are

$$\lim_{x \rightarrow \infty} x^k e^{-x} = 0$$

For any k

$$\lim_{x \rightarrow 0} x^k \ln x = 0$$

For any $k \neq 0$

Limits can be manipulated in order to use these two rules

Basic Skills

Q1. Explain what makes an integral improper

Q2. Determine if the improper integral $\int_0^1 \frac{1}{\sqrt{x}} dx$ exists and if so what the value is

Q3. Write the value of the following limits

a) $\lim_{x \rightarrow \infty} x^2 e^{-x}$

b) $\lim_{x \rightarrow 0} x^5 \ln x$

c) $\lim_{x \rightarrow \infty} \frac{3x^5}{e^x}$

d) $\lim_{x \rightarrow 0} \frac{\ln x}{2\sqrt{x}}$

Q4. Show that $\lim_{x \rightarrow \infty} 2x^2 \sqrt{x} e^{-2x} = 0$ by using the substitution $y = 2x$

Competency Skills

Q5. Prove that $\lim_{x \rightarrow \infty} x^k e^{-x} = 0$ by using

a) The Maclaurin's series for $e^x = 1 + x + \frac{x^2}{2} + \dots + \frac{x^k}{k!} + \dots$

b) L'hospital's rule $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}$

Q6. Prove that $\lim_{x \rightarrow 0} x^k \ln x = 0$ by using the substitution $x = e^{-\frac{y}{k}}$

Q7. Show that $\lim_{x \rightarrow 5} (x - 5) \ln(x - 5) = 0$

Q8. Determine the following improper integrals

a) $\int_0^\infty x e^{-x} dx$ **b)** $\int_0^\infty x e^x dx$ **c)** $\int_0^5 x^3 \ln x dx$

Advanced Skills

Q9. Find $\lim_{x \rightarrow 0} \left(1 - \frac{a}{(1+x)^3}\right) \ln x$ explaining the different cases depending on the real constant a

Q10. Find $\int_2^{3 \ln(x-2)} \frac{1}{x^2} dx$

Q11. For the function $f(x) = x^x$

a) Show that $f(x) = e^{x \ln x}$

b) Sketch the graph indicating any important behaviours

c) Show that $\lim_{x \rightarrow 0} x^x = 1$

Extension

Q12. Find $\int_0^1 x^x dx$

Hint: Use the Maclaurin's series for e^x